1) Project Name

Global Climate Change Tracking: A Comprehensive Exploratory Data Analysis of Historical Temperatures for SDG 13

2) Overview

This project aims to advance global climate monitoring by leveraging a comprehensive Exploratory Data Analysis (EDA) of historical temperature records to directly support Sustainable Development Goal (SDG) 13: Climate Action. The core idea is to transform raw, multi-layered spatiotemporal data from Berkeley Earth into a high-fidelity, machine-learning-ready temporal dataset. By implementing advanced data cleaning and aggregation techniques, we seek to establish a definitive climate baseline that distinguishes permanent, human-driven warming trends from the natural volatility of the Earth's climate system.

To ensure the highest analytical accuracy, we aim to isolate long-term anthropogenic shifts while filtering out short-term localized "noise" and extreme natural anomalies. Achieving this balance requires identifying and neutralizing significant disruptions, such as volcanic-induced stratospheric water vapor (e.g., the Hunga Tonga event), which can temporarily distort climate baselines. By applying systematic Data Aggregation across historical records—addressing missing values (NaNs) from the pre-19th century—we ensure that the output remains a robust architectural foundation for predictive AI modeling and "tipping point" detection.

This approach has potential applications in fields such as environmental risk assessment, urban planning, and climate policy development, where data integrity is critical for long-term forecasting. By providing a meticulously structured historical trajectory, highlighting the critical shift from a ~8.0°C baseline to over 9.5°C, we can significantly enhance the reliability of impact models for glacial melt and sea-level rise, fostering a more informed and proactive global response to the climate crisis.

3) Background

Climate change and global warming stand as the most urgent threats to sustainable development and global ecosystems today, aligning directly with Sustainable Development Goal (SDG) 13: Climate Action. The exponential increase in carbon emissions, driven by human activities such as industrialization, has triggered irreversible shifts in the Earth's climate system. To comprehend the magnitude of this crisis and develop data-driven interventions, an analysis of historical climate data is essential.

When examining the "Global Land Temperature Change" data spanning from 1750 to 2015, We can clearly see the warming trajectory. While annual average temperatures exhibit natural volatility, the last 40 years exposes a severe shift in temperature trends. Average land temperatures, which hovered around 8.0°C to 8.2°C in the 1850s, began a relentless climb in the 20th century. By 2015, this trendline surpassed 9.5°C, marking an increase of nearly 2 degrees. While a 2-degree shift might seem negligible in daily weather patterns, in the context of the global climate average, it represents a staggering accumulation of trapped heat energy. Much like a 2-degree increase in the human body induces a severe fever, this planetary warming acts as a catalyst for extreme weather, glacial melt, and the destabilization of historical climate norms.

However, analyzing this is complicated by extreme natural anomalies that introduce significant "noise" into the data. For instance, massive volcanic events temporarily disrupt the climate baseline. While typical eruptions inject sulfur dioxide that temporarily cools the atmosphere, unprecedented

underwater eruptions-such as the Hunga Tonga-Hunga Ha'apai Volcano Blast-can blast immense volumes of water vapor into the stratosphere, acting as a potent, but temporary, greenhouse gas. Differentiating between these natural spikes and the permanent, human-driven warming trend requires robust data processing capabilities. To forecast future climate scenarios, it is critical to systematically clean and aggregate comprehensive historical records. By resolving historical missing values (NaNs) from the pre-19th century and filtering out short-term localized noise, we can establish a high-quality dataset. This Exploratory Data Analysis (EDA) is the foundational step required before deploying predictive AI models to forecast future climate scenarios and tip points accurately.

4) Key Objectives / Business Objectives

a) Research Questions:

1. What is the definitive, long-term mathematical trend in average global land temperatures from the 18th century to the present day?
2. How do we deal with all the missing information (NaNs) from the 1700s and 1800s without messing up our final trendline?
3. What actual, useful lessons can we pull from this data to help policymakers fight climate change?
4. Why is forecasting future temperature trends crucial?
5. What are the risks of ignoring these long-term warming trends?
6. Why is historical data necessary to accurately forecast future climates?
7. What role does machine learning play in achieving SDG 13 (Climate Action)?
8. How will a 2-degree temperature rise affect our daily lives and economy?

b) Key Steps:

1. Data Acquisition: Locate and import the multi-layered Berkeley Earth datasets, ensuring they align with the project's sustainability objectives.
2. Data Cleaning: Systematically clean and preprocess the dataset by identifying anomalies, standardizing date formats, and resolving missing values (NaNs) to ensure high data quality.
3. Exploratory Analysis: Perform descriptive statistics to understand data distribution, variance, and central tendencies across different time periods.
4. Data Aggregation: Group the spatiotemporal data by year and region to compute annual mean temperatures, effectively filtering out short-term noise.
5. Visualization: Visualize key trends and patterns using advanced plotting libraries to create compelling evidence of the dataset's support for the project's goals.

5) Methods and Workflow

a) Datasets:

The project utilizes a relational set of historical climate data provided by Berkeley Earth. The specific files integrated into this analysis include:

- `GlobalTemperatures.csv`: The primary dataset containing overall global land and ocean temperatures.
- `GlobalLandTemperaturesByCountry.csv`, `GlobalLandTemperaturesByCity.csv` & `GlobalLandTemperaturesByState.csv`: Granular datasets allowing for geographical segmentation and localized trend analysis.

- `GlobalLandTemperaturesByMajorCity.csv`: Targeted data for assessing the impact of urbanization and localized climate shifts.

b) Data Cleaning and Preprocessing:

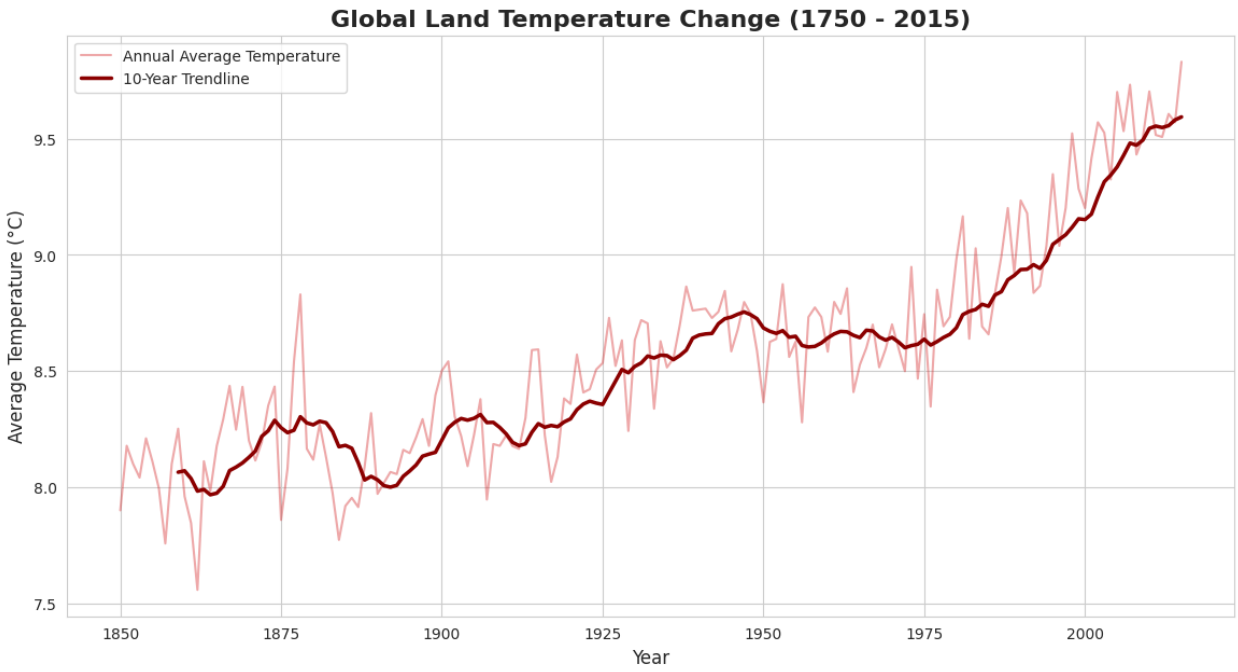
1. Conduct an initial audit of the datasets to map the frequency and distribution of missing values (NaN), particularly in records predating the 19th century.
2. Execute data filtering techniques, specifically utilizing the dropna method for critical temperature columns where imputation would introduce unacceptable historical bias.
3. Transform raw date strings into standard datetime objects, enabling accurate chronological indexing and temporal feature extraction.

c) Modelling and Analysis:

1. Implement aggregation functions to group the cleaned data by year, calculating the annual average global temperatures to smooth out seasonal and monthly variances.
2. Generate comprehensive descriptive statistics to evaluate the historical stability of the climate versus recent exponential changes.
3. Design and plot time-series visualizations, overlaying trend lines and moving averages (e.g., 10-year rolling averages) to visually validate the acceleration of global land temperatures up to the year 2015.

d) Deliverables:

1. A Machine-Learning-Ready Temporal Dataset: A meticulously cleaned and structured historical climate dataset. By systematically resolving missing values and filtering out natural anomalies (such as volcanic "noise"), this dataset serves as the robust foundational architecture required for deploying predictive AI models to forecast upcoming global temperature trends.
2. High-Fidelity Visualizations of Climate Trajectory: A suite of data visualizations clearly depicting the historical progression and severe acceleration of global warming (highlighting the critical shift from the ~8.0°C baseline to over 9.5°C).
3. A Strategic Framework for Impact and Risk Prediction: By establishing a definitive warming trendline, this analysis delivers the necessary groundwork to anticipate severe secondary consequences. The insights derived here pave the way for future studies to model and detect catastrophic environmental risks, such as accelerated glacial melt, imminent sea-level rise, and the irreversible destabilization of global ecosystems.
4. A Comprehensive Concept Note: This structured document explicitly summarizing the project choice, dataset selection, and EDA reasoning, strictly aligned with SDG 13 to support data-driven, actionable climate interventions.



İrem Akyol – 24061617
Murat Bilgin – 23061701
Barkın Girgin – 24061608
Eren Zeynal Toksoy – 22069502